

Towards Improving the Reliability of AI: What can Statistics Contribute?

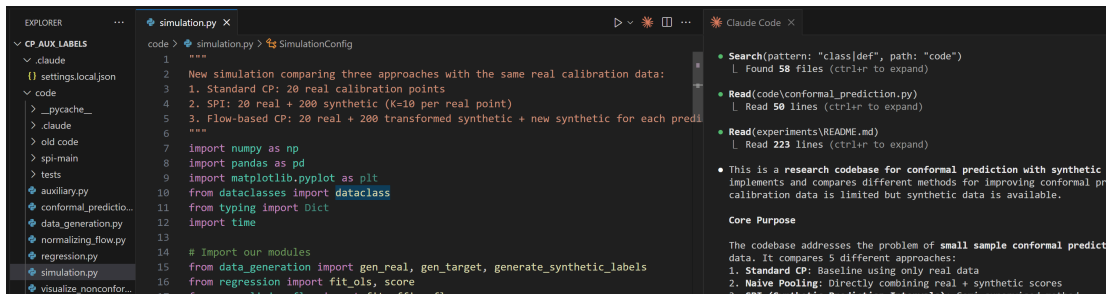
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AI is Great



The image shows a VS Code editor window with a file explorer on the left and a Claude Code sidebar on the right. The file explorer shows a project structure with files like `simulation.py`, `conformal_prediction.py`, and `data_generation.py`. The `simulation.py` file is open in the editor, showing a Python script that compares three approaches: Standard CP, SPI, and Flow-based CP. The Claude Code sidebar shows search results for the pattern `"class|def"` in the `code` path, finding 58 files. It also shows search results for `Read(code\conformal_prediction.py)` and `Read(experiments\README.md)`. The sidebar includes a section titled **Core Purpose** which states: "The codebase addresses the problem of small sample conformal prediction data. It compares 5 different approaches: 1. Standard CP: Baseline using only real data 2. Naive Pooling: Directly combining real + synthetic scores 3. SPI (Synthetic Prediction Interval): Semi-supervised method".

```
code > simulation.py > SimulationConfig
1 """
2 New simulation comparing three approaches with the same real calibration data:
3 1. Standard CP: 20 real calibration points
4 2. SPI: 20 real + 200 synthetic (K=10 per real point)
5 3. Flow-based CP: 20 real + 200 transformed synthetic + new synthetic for each predi
6 """
7 import numpy as np
8 import pandas as pd
9 import matplotlib.pyplot as plt
10 from dataclasses import dataclass
11 from typing import Dict
12 import time
13
14 # Import our modules
15 from data_generation import gen_real, gen_target, generate_synthetic_labels
16 from regression import fit_ols, score
```

Search(pattern: "class|def", path: "code")
L Found 58 files (ctrl+r to expand)

Read(code\conformal_prediction.py)
L Read 50 lines (ctrl+r to expand)

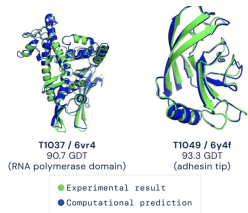
Read(experiments\README.md)
L Read 223 lines (ctrl+r to expand)

• This is a research codebase for conformal prediction with synthetic implements and compares different methods for improving conformal pr calibration data is limited but synthetic data is available.

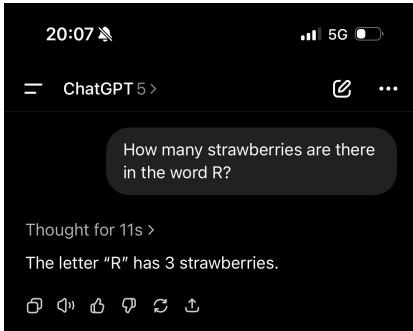
Core Purpose

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But also: AI is Terrible



The firm found that AI-assisted developers produced three to four times more code than their unassisted peers, but also generated ten times more security issues.



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- ▶ There is **no algorithm** to guarantee that a specific behavior occurs.
 - ▶ Important exception: When we have a perfect verifier.
- ▶ More generally, we can aim for probabilistic guarantees.

Directions where probabilistic and statistical thinking are especially promising

- ▶ Improving and changing behavior via high-probability wrappers
- ▶ Diagnostics and uncertainty quantification
- ▶ Evaluation
- ▶ Interventions and experiment design for steering and interpretation.
- ▶ Others: preference modelling, ...

Statistical Methods in Generative AI

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[Home](#) / [A-Z Publications](#) / [Annual Review of Statistics and Its Application](#)

Annual Review of Statistics and Its Application

From the review paper: Improving and changing behavior

Technique	Type	Examples
Change output	Additional output type	Highlight parts of output (Sun et al. 2022, Vasconcelos et al. 2025)
		Abstain from generation when a risk score is high (Farquhar et al. 2024, Yadkori et al. 2024)
		Add “Everything Else” as a possible answer (Noorani et al. 2025)
	Set of outputs	Construct prediction interval for each output coordinate (Horwitz & Hoshen 2022, Teneggi et al. 2023)
		Generate set of outputs (Quach et al. 2024, Gui et al. 2024, Nag et al. 2025)
	Trimmed output	Delete parts of output until correctness is achieved (Khakhar et al. 2023, Mohri & Hashimoto 2024)
		Find small parent set of possible outputs in a directed acyclic graph (Zhang et al. 2024)
	Regenerated output	Reformulate output until it is appropriately correct and specific (Jiang et al. 2025)
	Task-specific output	Train model to improve performance in downstream task (Band et al. 2024)
Construct prediction intervals for latent variables of a generated output (Sankaranarayanan et al. 2022)		
Interactively ask questions that maximize the informativeness of the answers (Chan et al. 2025)		
Change input	Set of inputs	Retrieve sets of documents in RAG (Li et al. 2024)
		Select prompts that control risk (Zollo et al. 2024)
Change other algorithm settings	—	Accelerate generation by early exit (Schuster et al. 2021, 2022, Jazbec et al. 2024)
		Reduce ambiguity by seeking additional input (Ren et al. 2023, 2024)
		Control a “size” component of the sampling mechanism (Ravfogel et al. 2023, Deutschmann et al. 2024, Ulmer et al. 2024)
		Switch between models when risk score is high (Overman & Bayati 2025)

Diagnostics and uncertainty quantification: rank-calibration

x: On September 28th, NASA announced that what had been detected on Mars?

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y: Possible signs of life

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- ▶ Which one is more useful?

Our answer: the uncertainty measure should reflect model accuracy (**rank-calibration**)

Uncertainty in Language Models: Assessment through Rank-Calibration

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Illustrating rank-calibration

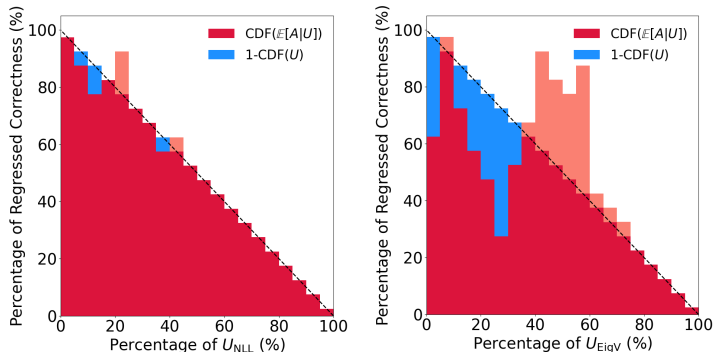


Figure: U_{NLL} (negative log-likelihood) and U_{EigV} , for the GPT-3.5-turbo model on the TriviaQA benchmark.